

Python Exercise (4 Hours)

1. Python program to check whether the given integer is a multiple of 5 and 7, and 10, 56.

Note: Input should be taken from the terminal.

2. Python program to find the factorial of a number using recursion.

Note: Input should be taken from the terminal.

3. Consider the following example of a 3x4 matrix implemented as a list of 3 lists of length.

For example:

Input:

```
[  
    [1, 2, 3, 4],  
    [5, 6, 7, 8],  
    [9, 10, 11, 12],  
]
```

Output:

```
[[1, 5, 9], [2, 6, 10], [3, 7, 11], [4, 8, 12]]
```

4. Find the even elements from an array. Respectively multiply this element in the array.

For example:

Input:

```
[1, 2, 3, 4, 5]
```

Output

```
[4,8]
```

Note: Need one line code, don't use for loop and if statement.

Input should be taken from the terminal

5. Write a program.

For example:

Input

[56.2, float('NaN'), 51.7, 55.3, 52.5, float('NaN'), 47.8]

Output

[56.2, 51.7, 55.3, 52.5, 47.8]

Note: No need to input

6. Flatten a list using list comprehension and for a loop.

For example:

Input:

[[1,2,3], [4,5,6], [7,8,9]]

Output:

[1, 2, 3, 4, 5, 6, 7, 8, 9]

Note: No need to input.

7. To read the last element from each row and sum of the all last elements.

For example:

Input

[[8, 14, -6], [12,7,4], [-11,3,21]]

Output

19

Note: Input should be taken from the terminal.